

MindEase Backend + SQL Task List

For Teammate 2

Backend + SQL Task List For MindEase

1. Database Setup

- Use SQLite for now: mindease.db
- Maintain tables: chat_history, moods, and optional emergency_contacts.

2. Chat History Backend

- Store every user and assistant message in chat_history.
- Fields: id, session_id, role, content, created_at.
- Fetch last 6 messages by session_id for chatbot memory.

3. Mood Tracking API

- Create and maintain POST /api/v1/mood.
- Save mood score from 1 to 10.
- Save optional note.
- Validate score before inserting into DB.

4. Mood Analytics API

- Create and maintain GET /api/v1/analytics/{session_id}.
- Return dates, scores, average mood, highest score, and lowest score.
- Support days query parameter.

5. SQL Table Improvements

- Add constraints: session_id NOT NULL, score CHECK score BETWEEN 1 AND 10.
- Add role check for user or assistant.
- Add indexes on session_id in chat_history and moods.
- Add created_at index for sorting and filtering.

6. Emergency Contact Storage

- Create table: id, session_id, name, telegram_chat_id, created_at.
- Add API to save or update emergency contact.
- Add API to get emergency contact by session.

7. Backend Code Cleanup

- Move repeated DB connection logic into app/core/database.py.
- Avoid creating tables inside route files if possible.
- Keep routes focused only on API logic.

8. Error Handling

- Return clear errors for missing session, invalid mood score, no mood data, and DB failure.
- Use proper HTTP status codes.

9. Testing

- Test /health, /api/v1/chat, /api/v1/mood, and /api/v1/analytics/{session_id}.

- Verify data is actually saved in SQLite.

10. Optional Upgrade

- If deployment needs a stronger DB, migrate from SQLite to PostgreSQL later.
- Add SQLAlchemy models and Alembic migrations.

Short version:

You handle backend + SQL: database schema, chat history storage, mood tracking APIs, mood analytics API, emergency contact table/API, DB cleanup, indexes/constraints, and API testing. Current DB is SQLite mindease.db; main tables are chat_history and moods.